Stage 1: Expanded Mathematical Formulation Notes

Free-Boundary-Aware Risk Surfaces for American Options

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Abstract

This teaching-oriented report develops the American-option payoff obstacle, time-to-maturity formulation, variational inequality, free boundary, and continuation premium step by step. It supplements rather than replaces the concise Stage 1 report.

Contents

1 Stage 1: Mathematical Foundations **2**

1.1 Purpose . . . 2

1.2 Method . . . 2

1.3 Mathematical Formulation . . . 2

1.3.1 Time to maturity . . . 2

1.3.2 Risk-neutral dynamics and the Black–Scholes operator . . . 3

1.3.3 From a PDE to a variational inequality . . . 3

1.3.4 Exercise region, continuation region, and free boundary . . . 3

1.3.5 Continuation premium . . . 4

1.4 Implementation Details . . . 4

1.5 Validation and Diagnostics . . . 4

1.6 Results . . . 5

1.7 Failure Modes and Limitations . . . 5

1.8 Conclusion and Next Step . . . 5

1 Stage 1: Mathematical Foundations

1.1 Purpose

This stage defines the financial problem and the notation used throughout the project. The objective is not to introduce a new option-pricing theory. It is to explain why early exercise changes a familiar Black–Scholes equation into an inequality problem with an unknown boundary.

1.2 Method

Let $S > 0$ denote the current asset price, $K > 0$ the strike, T the maturity date, $t \in [0, T]$ calendar time, r the continuously compounded risk-free rate, q the continuous dividend yield, and $\sigma > 0$ the volatility. The project studies American puts and American calls. Their immediate-exercise payoffs are

$$\Phi_{\text{put}}(S) = \max(K - S, 0), \qquad \Phi_{\text{call}}(S) = \max(S - K, 0).$$

The American value $V(S, t)$ is the best discounted expected payoff available when the holder may choose an exercise time θ between t and T:

$$V(S,t) = \sup_{\theta \in [t,T]} \mathbb{E}^{\mathbb{Q}}\left[e^{-r(\theta - t)} \Phi(S_\theta) \,\middle|\, S_t = S \right].$$

The expectation is under the risk-neutral measure $\mathbb{Q}$.

> **Reader intuition.** A European holder has one allowed exercise date. An American holder repeatedly compares two actions: exercise now for $\Phi(S)$, or retain the contract. The option value must therefore be at least the immediate payoff.

1.3 Mathematical Formulation

1.3.1 Time to maturity

The numerical solver uses

$$\tau = T - t, \qquad U(S, \tau) = V(S, T - \tau).$$

Thus $\tau = 0$ is maturity and $\tau = T$ is the initial valuation date. This change converts a terminal-value problem in t into an initial-value problem in τ:

$$U(S, 0) = \Phi(S).$$

> **Common confusion.** Increasing t moves forward in calendar time, but increasing τ moves away from maturity. The code begins with the payoff at $\tau = 0$ and advances through $\tau_1, \tau_2, \ldots, \tau_N = T$.

1.3.2 Risk-neutral dynamics and the Black–Scholes operator

Under the model used in the project,

$$dS_t = (r - q)S_t\,dt + \sigma S_t\,dW_t.$$

In a region where exercise is not optimal, standard replication and risk-neutral valuation give

$$U_\tau = \mathcal{L}U,$$

where

$$\mathcal{L}U = \frac{1}{2}\sigma^2 S^2 U_{SS} + (r - q)SU_S - rU.$$

The three terms have separate meanings:

- $\frac{1}{2}\sigma^2 S^2 U_{SS}$ is the diffusion or curvature term;
- $(r - q)SU_S$ is the risk-neutral drift term;
- $-rU$ is the discount term.

1.3.3 From a PDE to a variational inequality

Immediate exercise is always available, so

$$U(S, \tau) \geq \Phi(S).$$

If $U(S, \tau) > \Phi(S)$, retaining the option is more valuable and the continuation PDE applies:

$$U_\tau - \mathcal{L}U = 0.$$

If $U(S, \tau) = \Phi(S)$, the holder exercises. At such a point the PDE equality need not hold, but delaying exercise cannot produce a locally better value. The combined problem is

$$U(S, \tau) \geq \Phi(S), \qquad U_\tau - \mathcal{L}U \geq 0, \qquad \big(U - \Phi\big)\big(U_\tau - \mathcal{L}U\big) = 0.$$

> **Reader intuition.** The product condition says that at least one factor is zero. In continuation, $U - \Phi > 0$, so the PDE residual must be zero. In exercise, $U - \Phi = 0$, while the residual may be positive. This is why American pricing is an inequality or complementarity problem rather than only a linear PDE solve.

1.3.4 Exercise region, continuation region, and free boundary

Define

$$\mathcal{E} = \{(S, \tau) : U(S, \tau) = \Phi(S)\}, \qquad \mathcal{C} = \{(S, \tau) : U(S, \tau) > \Phi(S)\}.$$

The interface between $\mathcal{E}$ and $\mathcal{C}$ is the free boundary $B(\tau)$. It is "free" because its location is not prescribed as an input; it must be discovered together with the option value.

For a put, exercise generally occupies the low-spot side:

$$S \lesssim B_{\text{put}}(\tau) \quad \Longrightarrow \quad \text{exercise.}$$

For a dividend-paying call, exercise—when it occurs—generally occupies the high-spot side:

$$S \gtrsim B_{\text{call}}(\tau) \quad \Longrightarrow \quad \text{exercise.}$$

A no-dividend American call is retained as a control because, under the standard assumptions, early exercise is not optimal.

1.3.5 Continuation premium

The continuation premium is

$$C(S,\tau) = U(S,\tau) - \Phi(S).$$

The obstacle condition immediately gives

$$C(S,\tau) \geq 0.$$

Moreover,

$$C = 0 \quad \text{in the exercise region,} \qquad C > 0 \quad \text{in the continuation region.}$$

> **Reader intuition.** C is the extra value of keeping the option alive instead of exercising immediately. It is therefore both a financial quantity and a useful numerical indicator of the exercise boundary.

> **Common confusion.** Nonnegative continuation premium is only the payoff-obstacle requirement. It does not by itself guarantee monotonicity, convexity, a correct free boundary, or accurate Delta and Gamma.

1.4 Implementation Details

The solver stores spot along one array axis and time to maturity along the other. Reports use normalized quantities such as S/K, U/K, and B/K where useful, but the mathematical obstacle remains $U \geq \Phi$. Later neural features include

$$x = \big(\log(S/K), \tau/T, r, q, \sigma, T, \mathbf{1}_{\text{call}}\big).$$

1.5 Validation and Diagnostics

Before discretization, the notation is checked for the direction of time, the sign of $U_\tau - \mathcal{L}U$, put/call payoff orientation, and the units of $\Delta = U_S$ and $\Gamma = U_{SS}$. A sign error can yield stable-looking arrays while solving the wrong problem.

1.6 Results

Stage 1 produces a consistent mathematical contract. It introduces no new numerical results and makes no model-performance claim.

1.7 Failure Modes and Limitations

The formulation assumes the Black–Scholes diffusion with constant parameters. It does not include jumps, stochastic volatility, transaction costs, or market calibration. These omissions delimit the project rather than invalidate its within-model numerical study.

1.8 Conclusion and Next Step

The continuous problem is an obstacle-constrained PDE. Stage 2 converts its spatial operator into a tridiagonal matrix, applies Crank–Nicolson in time, and solves the resulting linear complementarity problem with PSOR.